

Unit 4: Image Restoration and Compression

4.2: Restoration Filters

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Overview of Restoration Filters

Purpose

Restoration filters are designed to remove or reduce noise and blur from degraded images while preserving important image details.

Categories of Restoration Filters:

- ① **Mean Filters:** Work with pixel intensity averages
 - Arithmetic, Geometric, Harmonic, Contraharmonic
- ② **Order Statistics Filters:** Based on ranking pixel values
 - Median, Min, Max, Midpoint, Alpha-trimmed mean
- ③ **Frequency Domain Filters:** Band pass and Band reject
 - Ideal, Butterworth, Gaussian

Spatial vs. Frequency Domain Filtering

Spatial Domain

- Operates directly on pixels
- Uses convolution with masks/kernels
- Mean and order statistics filters
- Computationally simpler
- Better for localized operations

Frequency Domain

- Operates on Fourier transform
- Uses multiplication in frequency
- Band pass/reject filters
- Better for periodic noise
- Global operations

Key Concept

Convolution in spatial domain $\equiv$ Multiplication in frequency domain

Mean Filters - Introduction

General Concept:

- Work by replacing each pixel with some form of average of its neighborhood
- Effective for removing random noise
- Simple to implement
- Trade-off between noise reduction and detail preservation

General Notation:

- S_{xy} : Set of coordinates in a neighborhood centered at (x, y)
- $g(x, y)$: Restored image
- $f(s, t)$: Degraded image at coordinates (s, t)
- Neighborhood typically 3×3 , 5×5 , 7×7 , etc.

Arithmetic Mean Filter

Formula

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} g(s, t)$$

where $m \times n$ is the size of the neighborhood S_{xy} .

Characteristics:

- Simplest of all mean filters
- Computes the average intensity of neighborhood pixels
- Each pixel given equal weight
- Reduces noise by smoothing

Advantages:

- Simple and fast computation
- Effective for Gaussian noise
- Good for uniform noise

Disadvantages:

- Causes blurring (loss of sharp edges)
- Not effective for salt-and-pepper noise

Arithmetic Mean Filter - Example

Example Calculation:

Consider a 3×3 neighborhood:

$$\begin{bmatrix} 10 & 15 & 20 \\ 12 & 18 & 22 \\ 14 & 16 & 24 \end{bmatrix}$$

Arithmetic Mean:

$$\hat{f}(x, y) = \frac{1}{9}(10 + 15 + 20 + 12 + 18 + 22 + 14 + 16 + 24) = \frac{151}{9} \approx 16.78$$

Application Notes:

- Larger neighborhoods: More smoothing, more blur
- Smaller neighborhoods: Less smoothing, preserves more detail
- Commonly used as preprocessing step

Geometric Mean Filter

Formula

$$\hat{f}(x, y) = \left[\prod_{(s,t) \in S_{xy}} g(s, t) \right]^{1/mn}$$

where the product is over all pixels in the $m \times n$ neighborhood.

Characteristics:

- Uses geometric mean instead of arithmetic mean
- Product of all pixel values raised to power $1/mn$
- Tends to lose less detail compared to arithmetic mean

Advantages:

- Better detail preservation than arithmetic mean
- Good smoothing properties
- Effective for Gaussian noise

Geometric Mean Filter - Example

Example Calculation:

Using the same 3×3 neighborhood:

$$\begin{bmatrix} 10 & 15 & 20 \\ 12 & 18 & 22 \\ 14 & 16 & 24 \end{bmatrix}$$

Geometric Mean:

$$\begin{aligned} \hat{f}(x, y) &= (10 \times 15 \times 20 \times 12 \times 18 \times 22 \times 14 \times 16 \times 24)^{1/9} \\ &= (1,769,472,000)^{1/9} \\ &\approx 16.38 \end{aligned}$$

Note

Geometric mean $\leq$ Arithmetic mean (by AM-GM inequality)

Harmonic Mean Filter

Formula

$$\hat{f}(x, y) = \frac{mn}{\sum_{(s,t) \in S_{xy}} \frac{1}{g(s, t)}}$$

Characteristics:

- Based on reciprocals of pixel intensities
- Works well for salt noise (bright spots)
- Fails for pepper noise (dark spots)
- Cannot handle zero-valued pixels (division by zero)

Advantages:

- Excellent for salt noise removal
- Better than arithmetic mean for certain noise types

Disadvantages:

- Cannot handle zero or negative pixel values
- Not suitable for pepper noise
- May produce artifacts

Harmonic Mean Filter - Example

Example Calculation:

Using the same 3×3 neighborhood:

$$\begin{bmatrix} 10 & 15 & 20 \\ 12 & 18 & 22 \\ 14 & 16 & 24 \end{bmatrix}$$

Harmonic Mean:

$$\begin{aligned} \hat{f}(x, y) &= \frac{9}{\frac{1}{10} + \frac{1}{15} + \frac{1}{20} + \frac{1}{12} + \frac{1}{18} + \frac{1}{22} + \frac{1}{14} + \frac{1}{16} + \frac{1}{24}} \\ &= \frac{9}{0.5637} \\ &\approx 15.96 \end{aligned}$$

Order Relationship:

$$\text{Harmonic Mean} \leq \text{Geometric Mean} \leq \text{Arithmetic Mean}$$

Contraharmonic Mean Filter

Formula

$$\hat{f}(x, y) = \frac{\sum_{(s,t) \in S_{xy}} g(s, t)^{Q+1}}{\sum_{(s,t) \in S_{xy}} g(s, t)^Q}$$

where Q is the order of the filter.

Order Parameter Q :

- $Q > 0$ (**Positive**): Eliminates pepper noise (dark spots)
- $Q < 0$ (**Negative**): Eliminates salt noise (bright spots)
- $Q = 0$: Reduces to arithmetic mean filter
- $Q = -1$: Reduces to harmonic mean filter

Contraharmonic Mean Filter - Properties

Characteristics:

- Most versatile of mean filters
- Behavior controlled by order Q
- Can handle both salt and pepper noise (but not simultaneously)
- Reduces to other mean filters for special values of Q

Advantages:

- Flexible - adjustable parameter Q
- Effective for impulse noise
- Can be tuned for specific noise types

Disadvantages:

- Cannot eliminate salt and pepper noise simultaneously
- Must choose Q carefully
- Wrong choice of Q can make noise worse
- Computationally expensive for high $|Q|$

Contraharmonic Mean Filter - Examples

For $Q = 1$:

$$\hat{f}(x, y) = \frac{\sum g(s, t)^2}{\sum g(s, t)}$$

Good for pepper noise removal.

For $Q = -1$:

$$\hat{f}(x, y) = \frac{\sum g(s, t)}{\sum \frac{1}{g(s, t)}} = \text{Harmonic Mean}$$

For $Q = 0$:

$$\hat{f}(x, y) = \frac{\sum g(s, t)}{mn} = \text{Arithmetic Mean}$$

Important

The filter must be used with the correct sign of Q for the noise type, otherwise it will intensify the noise!

Comparison of Mean Filters

Filter	Best For	Drawbacks
Arithmetic Mean	Gaussian noise, uniform noise	Blurs edges significantly
Geometric Mean	Gaussian noise with better detail preservation	Computational cost, sensitive to zeros
Harmonic Mean	Salt noise (bright spots)	Cannot handle pepper noise or zeros
Contraharmonic Mean	Salt or pepper noise (choose Q sign)	Cannot handle both simultaneously

General Guidelines:

- Use arithmetic/geometric for Gaussian noise
- Use harmonic/contraharmonic for impulse noise
- For mixed noise, use order statistics filters (next section)

Order Statistics Filters - Introduction

Definition

Order statistics filters are nonlinear spatial filters whose response is based on ordering (ranking) the pixels in the neighborhood and then replacing the center pixel with a value determined from the ranking result.

Key Concept:

- 1 Sort all pixels in neighborhood by intensity
- 2 Select a pixel based on its rank position
- 3 Replace center pixel with selected value

Advantages over Mean Filters:

- Better edge preservation
- Effective for impulse noise (salt-and-pepper)
- Less blurring
- Nonlinear - can preserve features better

Median Filter

Formula

$$\hat{f}(x, y) = \text{median}_{(s,t) \in S_{xy}} \{g(s, t)\}$$

The median is the middle value in the sorted list of pixel intensities.

Algorithm:

- 1 Extract neighborhood pixels
- 2 Sort pixels in ascending order
- 3 Select the middle value (or average of two middle values for even count)
- 4 Replace center pixel with median value

Characteristics:

- Most popular order statistics filter
- Excellent for salt-and-pepper (impulse) noise
- Preserves edges much better than mean filters

Median Filter - Properties and Example

Advantages:

- Excellent for salt-and-pepper noise removal
- Preserves edges and sharp details
- Less sensitive to outliers than mean filters
- Does not create new pixel values (uses existing values)

Disadvantages:

- Computationally more expensive (requires sorting)
- May remove thin lines and small details
- Less effective for Gaussian noise

Example: Neighborhood pixels: $\{10, 15, 20, 12, 255, 22, 14, 16, 24\}$ (one noisy pixel = 255)

Sorted: $\{10, 12, 14, 15, 16, 20, 22, 24, 255\}$

Median = 16 (middle value) - *noise removed!*

Max Filter

Formula

$$\hat{f}(x, y) = \max_{(s, t) \in S_{xy}} \{g(s, t)\}$$

Selects the maximum pixel value in the neighborhood.

Characteristics:

- Finds the brightest pixel in neighborhood
- Useful for finding the brightest spots
- Reduces pepper noise (dark spots)
- Tends to brighten the image

Applications:

- Removing pepper noise
- Dilating bright objects
- Enhancing bright features

Example: Pixels: {10, 15, 20, 12, 0, 22, 14, 16, 24} (pepper noise = 0)

Formula

$$\hat{f}(x, y) = \min_{(s, t) \in S_{xy}} \{g(s, t)\}$$

Selects the minimum pixel value in the neighborhood.

Characteristics:

- Finds the darkest pixel in neighborhood
- Useful for finding the darkest spots
- Reduces salt noise (bright spots)
- Tends to darken the image

Applications:

- Removing salt noise
- Eroding bright objects
- Enhancing dark features

Example: Pixels: $\{10, 15, 20, 12, 255, 22, 14, 16, 24\}$ (salt noise = 255)

Midpoint Filter

Formula

$$\hat{f}(x, y) = \frac{1}{2} \left[\max_{(s,t) \in S_{xy}} \{g(s, t)\} + \min_{(s,t) \in S_{xy}} \{g(s, t)\} \right]$$

The midpoint is the average of the maximum and minimum values in the neighborhood.

Characteristics:

- Combines max and min filters
- Computes average of extreme values
- Works well for uniformly distributed noise
- Also effective for Gaussian noise

Example: Pixels: {10, 15, 20, 12, 18, 22, 14, 16, 24}

Max = 24, Min = 10

Midpoint = $\frac{24+10}{2} = 17$

Midpoint Filter - Properties

Advantages:

- Good for randomly distributed noise (uniform, Gaussian)
- Simpler computation than median (no sorting needed)
- Works well when noise is symmetrically distributed
- Faster than median filter

Disadvantages:

- Not as effective as median for impulse noise
- Sensitive to extreme outliers
- May not preserve edges as well as median

Best Applications:

- Images with uniform noise
- Images with Gaussian noise
- When computational speed is important

Alpha-Trimmed Mean Filter

Formula

$$\hat{f}(x, y) = \frac{1}{mn - d} \sum_{(s,t) \in S_{xy}} g_r(s, t)$$

where g_r represents the remaining pixels after deleting the $d/2$ lowest and $d/2$ highest intensity values from the neighborhood, and d is the number of pixels trimmed (must be even).

Algorithm:

- 1 Sort neighborhood pixels
- 2 Delete $d/2$ smallest values
- 3 Delete $d/2$ largest values
- 4 Compute arithmetic mean of remaining pixels

Alpha-Trimmed Mean Filter - Properties

Special Cases:

- $d = 0$: Reduces to arithmetic mean filter
- $d = mn - 1$: Reduces to median filter

Characteristics:

- Combines benefits of mean and median filters
- Trims extreme values before averaging
- Useful for mixed noise (Gaussian + impulse)
- More robust than arithmetic mean

Advantages:

- Excellent for images with multiple noise types
- Removes outliers while averaging
- Better edge preservation than arithmetic mean
- Flexible - adjustable parameter d

Alpha-Trimmed Mean Filter - Example

Example with $d = 4$:

Neighborhood (3×3 , 9 pixels): Original: $\{10, 15, 20, 12, 0, 22, 255, 16, 24\}$

Step 1: Sort Sorted: $\{0, 10, 12, 15, 16, 20, 22, 24, 255\}$

Step 2: Trim $d/2 = 2$ from each end After trimming: $\{12, 15, 16, 20, 22\}$

Step 3: Compute mean

$$\hat{f}(x, y) = \frac{12 + 15 + 16 + 20 + 22}{5} = \frac{85}{5} = 17$$

Both extreme noisy values (0 and 255) were removed!

Comparison of Order Statistics Filters

Filter	Best For	Key Feature
Median	Salt-and-pepper noise	Best edge preservation
Max	Pepper noise (dark spots)	Brightens image
Min	Salt noise (bright spots)	Darkens image
Midpoint	Uniform and Gaussian noise	Fast, uses extremes only
Alpha-trimmed	Mixed noise types	Combines mean and median benefits

Selection Guidelines:

- Pure impulse noise → Median filter
- Mixed noise → Alpha-trimmed mean filter
- Salt only → Min filter
- Pepper only → Max filter

Frequency Domain Filtering - Introduction

Basic Concept

Frequency domain filtering involves transforming the image to frequency domain using Fourier Transform, applying a filter, and transforming back to spatial domain.

Process:

$$\text{Image } f(x, y) \xrightarrow{\text{FFT}} F(u, v) \xrightarrow{\text{Filter } H(u, v)} G(u, v) \xrightarrow{\text{IFFT}} g(x, y)$$

Key Principle:

$$g(x, y) = \mathcal{F}^{-1}[H(u, v) \cdot F(u, v)]$$

where:

- $F(u, v)$ = Fourier transform of input image
- $H(u, v)$ = Frequency domain filter transfer function
- $G(u, v)$ = Filtered image in frequency domain
- $g(x, y)$ = Restored image in spatial domain

Frequency Domain Concepts

Frequency Components:

- **Low frequencies:** Smooth regions, slow intensity changes
- **High frequencies:** Edges, noise, fine details, rapid changes
- **DC component:** Center of spectrum (average intensity)

Distance from Origin:

$$D(u, v) = \sqrt{u^2 + v^2}$$

where (u, v) are frequency domain coordinates.

Filter Types by Frequency Response:

- **Low-pass:** Allows low frequencies (smoothing)
- **High-pass:** Allows high frequencies (sharpening)
- **Band-pass:** Allows a band of frequencies
- **Band-reject (Notch):** Rejects a band of frequencies

Band Pass Filters - Overview

Definition

Band pass filters allow frequencies within a certain range to pass while attenuating frequencies outside this range.

General Form:

$$H_{BP}(u, v) = 1 - H_{BR}(u, v)$$

where H_{BR} is the corresponding band reject filter.

Characteristics:

- Pass frequencies between D_0 (inner radius) and D_1 (outer radius)
- Useful for isolating specific frequency bands
- Can extract features at particular scales

Applications:

- Extracting features in specific frequency ranges
- Texture analysis
- Pattern recognition

Band Reject Filters - Overview

Definition

Band reject (notch) filters attenuate frequencies within a certain range while passing frequencies outside this range.

Purpose:

- Remove periodic noise (e.g., from scanning, electronic interference)
- Eliminate specific frequency components
- Preserve most image information while removing interference

Applications:

- Removing periodic noise patterns
- Eliminating interference from power lines (50/60 Hz)
- Removing scanning artifacts and Cleaning up images with regular noise patterns

Key Parameters:

- D_0 : Inner cutoff frequency (center of reject band)
- W : Width of the reject band

Ideal Band Reject Filter

Transfer Function

$$H(u, v) = \begin{cases} 0 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$$

where:

- $D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$ (distance from center)
- D_0 = center frequency of reject band
- W = width of the reject band

Characteristics:

- Sharp cutoff (binary: 0 or 1)
- Completely rejects frequencies in the band
- Completely passes frequencies outside the band

Ideal Band Reject Filter - Properties

Advantages:

- Perfect frequency selectivity
- Simple mathematical definition
- Complete rejection in stop band
- Easy to understand conceptually

Disadvantages:

- Causes severe ringing artifacts (Gibbs phenomenon)
- Not realizable in practice (infinite impulse response)
- Sharp transitions create spatial domain artifacts
- Poor performance in practice

Ideal Band Pass Filter:

$$H_{BP}(u, v) = \begin{cases} 1 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 0 & \text{otherwise} \end{cases}$$

Butterworth Band Reject Filter

Transfer Function

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v) \cdot W}{D^2(u, v) - D_0^2} \right]^{2n}}$$

where:

- $D(u, v)$ = distance from frequency domain origin
- D_0 = center (notch) frequency
- W = width of the reject band
- n = order of the filter

Characteristics:

- Smooth transition between pass and reject regions
- No ringing artifacts
- Controllable sharpness via order n
- Most commonly used in practice

Butterworth Band Reject Filter - Properties

Filter Order n :

- $n = 1$: Gradual transition (first-order)
- $n = 2$: Moderate transition (second-order)
- $n \rightarrow \infty$: Approaches ideal filter
- Higher n : Sharper cutoff, more ringing

Advantages:

- Smooth frequency response (no ringing)
- Adjustable sharpness through order n
- Good compromise between ideal and Gaussian and Widely used in practice

Butterworth Band Pass Filter:

$$H_{BP}(u, v) = 1 - H_{BR}(u, v)$$

or directly:

$$H_{BP}(u, v) = \frac{1}{1 + \left[\frac{D^2(u, v) - D_0^2}{D(u, v) \cdot W} \right]^{2n}}$$

Gaussian Band Reject Filter

Transfer Function

$$H(u, v) = 1 - e^{-\frac{[D^2(u, v) - D_0^2]^2}{2[D(u, v) \cdot W]^2}}$$

where:

- $D(u, v)$ = distance from origin
- D_0 = center frequency of reject band
- W = width parameter (controls band width)

Characteristics:

- Based on Gaussian function
- Very smooth transition (smoothest of all three)
- No ringing at all
- Gentle rolloff

Gaussian Band Reject Filter - Properties

Advantages:

- Absolutely no ringing artifacts
- Smoothest possible transition
- Minimum spatial domain distortion
- Gaussian in both spatial and frequency domains

Disadvantages:

- Less sharp cutoff than Butterworth or Ideal
- May not reject frequencies as completely
- Wider transition region
- Less selective than Butterworth

Gaussian Band Pass Filter:

$$H_{BP}(u, v) = e^{-\frac{[D^2(u, v) - D_0^2]^2}{2[D(u, v) \cdot W]^2}}$$

Note

For Gaussian, the band pass filter is simply the exponential term, not $1 - H_{BR}$.

Comparison of Band Reject Filters

Filter Type	Transition	Ringing	Selectivity
Ideal	Sharp (step function)	Severe	Highest
Butterworth	Smooth, adjustable with n	Minimal to none	Moderate
Gaussian	Smoothest	None	Lowest

Selection Guidelines:

- **Ideal:** Theoretical study only (not practical)
- **Butterworth:** Most commonly used - good balance
- **Gaussian:** When smoothness is critical, artifacts must be avoided

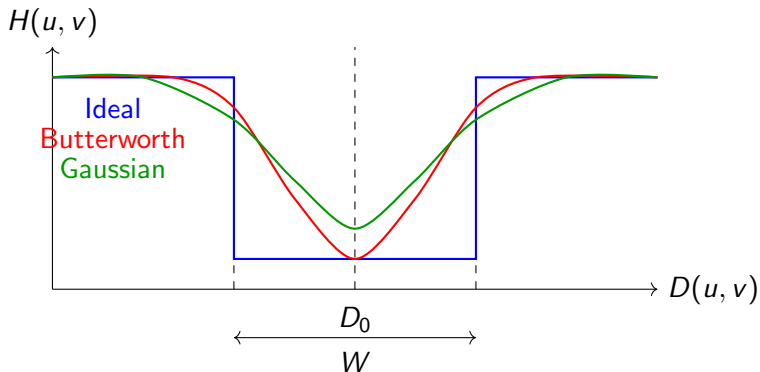
Trade-off:

Sharper Cutoff $\leftrightarrow$ More Ringing/Artifacts

Frequency Response Visualization

Comparative Frequency Responses:

For all three band reject filters centered at D_0 with width W :



Notice: Ideal has sharp discontinuities, Butterworth is smooth, Gaussian is smoothest.

Applications of Band Reject Filters

Common Applications:

① Periodic Noise Removal:

- Scanning line patterns
- Interference patterns from electronic equipment
- Regular grid patterns

② Specific Examples:

- Power line interference (50/60 Hz hum)
- Moiré patterns in scanned documents
- TV broadcast interference patterns
- Regular texture removal

③ Frequency Domain Analysis:

- Identify noise spikes in FFT
- Design notch filter at spike locations
- Apply filter and transform back

Practical Implementation Steps

Step-by-Step Process for Band Reject Filtering:

① Transform to Frequency Domain:

$$F(u, v) = \mathcal{F}\{f(x, y)\}$$

② Identify Noise Frequencies:

- Visualize magnitude spectrum
- Locate periodic noise spikes
- Determine D_0 and W

③ Design Filter:

- Choose filter type (Ideal/Butterworth/Gaussian)
- Set parameters (D_0 , W , n for Butterworth)

④ Apply Filter:

$$G(u, v) = H(u, v) \cdot F(u, v)$$

⑤ Transform Back:

$$g(x, y) = \mathcal{F}^{-1}\{G(u, v)\}$$

Summary - Mean Filters

Mean Filters Overview:

Filter	Formula	Best Use
Arithmetic Mean	$\frac{1}{mn} \sum g(s, t)$	Gaussian noise
Geometric Mean	$[\prod g(s, t)]^{1/mn}$	Gaussian with detail preservation
Harmonic Mean	$\frac{mn}{\sum 1/g(s, t)}$	Salt noise
Contraharmonic	$\frac{\sum g^{Q+1}}{\sum g^Q}$	Salt ($Q \geq 0$) or pepper ($Q < 0$)

Key Points:

- All cause some blurring
- Trade-off between noise reduction and detail loss
- Linear operations (except geometric)

Summary - Order Statistics Filters

Order Statistics Filters Overview:

Filter	Operation	Best Use
Median	Middle value of sorted list	Salt-and-pepper noise
Max	Maximum value	Pepper noise
Min	Minimum value	Salt noise
Midpoint	Average of max and min	Uniform/Gaussian noise
Alpha-trimmed	Mean after trimming extremes	Mixed noise

Key Points:

- Nonlinear operations
- Better edge preservation than mean filters
- Excellent for impulse noise

Summary - Frequency Domain Filters

Band Reject/Pass Filters Overview:

Filter Type	Characteristics	Application
Ideal	Sharp cutoff, severe ringing	Theoretical only
Butterworth	Smooth, adjustable order	General purpose, most common
Gaussian	Smoothest, no ringing	Minimum artifacts needed

Key Points:

- Used for periodic noise removal
- Band reject eliminates frequency bands
- Band pass isolates frequency bands
- Transform required: Spatial $\leftrightarrow$ Frequency

Overall Comparison and Guidelines

Filter Selection Guide:

- **Gaussian Noise:**
 - Spatial: Arithmetic or Geometric Mean
 - Frequency: Butterworth Low-pass
- **Impulse Noise (Salt-and-Pepper):**
 - Best: Median Filter
 - Alternative: Alpha-trimmed Mean
- **Salt Noise Only:**
 - Min or Harmonic Mean or Contraharmonic (Q_i0)
- **Pepper Noise Only:**
 - Max or Contraharmonic (Q_i0)
- **Periodic Noise:**
 - Band Reject Filter (Butterworth preferred)
- **Mixed Noise:**
 - Alpha-trimmed Mean Filter

Text Books:

- ① Rafael C. Gonzalez and Richard E. Woods, "*Digital Image Processing*", Pearson Edition, Latest Edition.

Reference Books:

- ② I. Pitas, "*Digital Image Processing Algorithms*", Prentice Hall, Latest Edition.
- ③ A. K. Jain, "*Fundamental of Digital Image Processing*", Prentice Hall of India Pvt. Ltd., Latest Edition.
- ④ K. Castleman, "*Digital Image Processing*", Prentice Hall of India Pvt. Ltd., Latest Edition.

Thank You!

Questions?